

KPIFlow Enterprise

In-House Goal Setting & Performance Tracking Portal

Problem Statement	In-House Goal Setting & Tracking Portal
Technology Stack	Python / Flask • SQLite • HTML / CSS / JS
Architecture	Monolithic MVC with SSE real-time sync
Roles Supported	Employee • Manager • Admin
Working URL	https://kpiforge.onrender.com
Repository	https://github.com/gautam-gowda/KPIForge

DEMO LOGIN CREDENTIALS

Role	Username			Password
Employee	employee1	employee2	employee3	pass123
Manager	manager			pass123
Admin	admin			pass123

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EXECUTIVE SUMMARY

KPIFlow Enterprise is a fully functional, web-based KPI governance portal built to address the shortcomings of spreadsheet-driven goal management in organisations. It digitises the complete employee performance lifecycle — from goal creation and manager approval through quarterly check-ins to executive analytics — across three distinct user roles.

The system implements every mandatory requirement specified in the problem statement (BRD Phases 1 and 2), including shared goal management, system-computed performance scoring, escalation monitoring, and exportable audit-ready reports. Real-time dashboard synchronisation via Server-Sent Events (SSE) ensures all stakeholders see live updates without manual refreshes.

20+ Features

All BRD requirements

Real-Time Sync

SSE + partial DOM

3 Role Dashboards

Distinct UI per role

Export Ready

CSV & Excel

6 UoM Types

Full scoring engine

Audit Trail

Every action logged

PROBLEM STATEMENT

Organisations relying on manual or fragmented goal-tracking methods face persistent challenges in alignment, visibility, and accountability. Spreadsheets, emails, and offline review cycles create blind spots — managers cannot monitor team progress in real time, employees lack clarity on how their work connects to organisational priorities, and HR teams are left piecing together data at appraisal time.

The problem statement requires a structured, digital **Goal Setting & Tracking Portal** that supports:

- **Phase 1** — Goal creation, weighted allocation, UoM assignment, and manager approval workflow
- **Phase 2** — Quarterly achievement tracking, performance score computation, and check-in comments
- **Governance** — Exportable reports, completion dashboards, and immutable audit trails
- **Bonus Modules** — Escalation monitoring, analytics, SSO, and notification integrations

SOLUTION OBJECTIVE

KPIFlow Enterprise is designed to function as an *internal enterprise SaaS prototype* — not a CRUD demo. The design priorities are:

1. **Completeness** — Every mandatory BRD requirement is implemented and validated
2. **Correctness** — Business rules (weightage limits, window enforcement, scoring formulae) are server-enforced, not client-only
3. **Security** — CSRF protection, hashed passwords, role-based access control on every route
4. **Real-Time Governance** — Live dashboard sync via SSE with partial DOM patching (no full-page reload)
5. **Audit Readiness** — Every state change is timestamped and persisted in an immutable audit log

BRD REQUIREMENT ALIGNMENT

4.1 Phase 1 — Goal Creation & Approval

Requirement	Implementation	Status
Employee goal creation interface	Form with all required fields	✓
Select Thrust Area & Goal Title/Description	6 areas; title + textarea	✓
Assign UoM (Numeric / % / Timeline / Zero)	All 6 types with UX hints	✓
Set Target & Weightage per goal	Numeric/date target; 10–100%	✓
Total weightage must equal 100%	Server-side enforced + flash	✓
Min 10% weightage per goal	Hard-validated in create_goal()	✓
Max 8 goals per employee	Hard limit enforced	✓
Manager L1 review — edit targets inline	Edit form per goal row	✓
Manager approve / return for rework	Approve / Rework / Reject flows	✓
Goals locked after approval	finalized flag; Admin can unlock	✓
Shared goals push to all employees	Manager pushes to 3 employees	✓
Recipients adjust weightage only	Title & target read-only in UI	✓
Achievement sync across shared goals	Synced in update_achievement()	✓

4.2 Phase 2 — Achievement Tracking & Check-ins

Requirement	Implementation	Status
Quarterly achievement entry	Per check-in window; date or numeric	✓
Status: Not Started / On Track / Completed	Start Goal / Mark Completed	✓
Manager check-in comment module	quarterly_review textarea per goal	✓
System-computed progress scores	Formula per UoM; 0–100% capped	✓
Quarterly window enforcement	Month-gated via get_active_window()	✓

4.3 Governance & Reporting

Requirement	Implementation	Status
CSV export (all fields, all employees)	/export_csv — 19 fields	✓
Excel export with formatting	openpyxl with header fill + alt rows	✓
Completion dashboard	Verified/total per employee	✓
Audit trail — who changed what, when	audit_log col; timestamped entries	✓
Three distinct user roles	Role decorators on every route	✓

4.4 Bonus Features Implemented

Module	Details	Status
Escalation Module (5.3)	3 escalation rules: no goals submitted, not finalised, no check-in entries; all window-aware	✓
Analytics Module (5.4)	6 Chart.js charts: approval distribution, employee avg score, thrust area breakdown, UoM breakdown, progress status, weightage allocation	✓

KEY INNOVATION HIGHLIGHTS

5.1 SSE Live Dashboard Sync

The dashboards stay current without full page reloads. A **Server-Sent Events** stream (/stream) polls the database fingerprint every 3 seconds. On detecting a change, it emits a typed JSON event. The client JavaScript calls a lightweight JSON API endpoint (/api/employee_goals or /api/manager_summary), then patches only the changed DOM elements with a blue highlight flash.

This approach solved a critical session-cookie loss bug that affected the original SSE implementation — browsers drop Set-Cookie headers on EventSource connections, causing session destruction on the next POST. The fix avoids full-page reloads entirely.

5.2 Multi-Type Scoring Engine

A single calculate_score(goal) function handles all six UoM types with appropriate edge case handling:

UoM Type	Formula
Numeric-Min / %-Min	$S = \min \left(\frac{A}{T} \times 100, 100 \right)$
Numeric-Max / %-Max	$S = \min \left(\frac{\bar{T}}{A} \times 100, 100 \right)$
Timeline	$S = 100$ if on time; $\max \left(0, 100 - \frac{d_{\text{late}}}{d_{\text{year}}} \times 100 \right)$ if late
Zero-based	$S = 100$ if $A = 0$; else $S = 0$

A = Actual Achievement, T = Target, S = Score (%), d_{late} = days overdue

5.3 Shared Goal Synchronisation

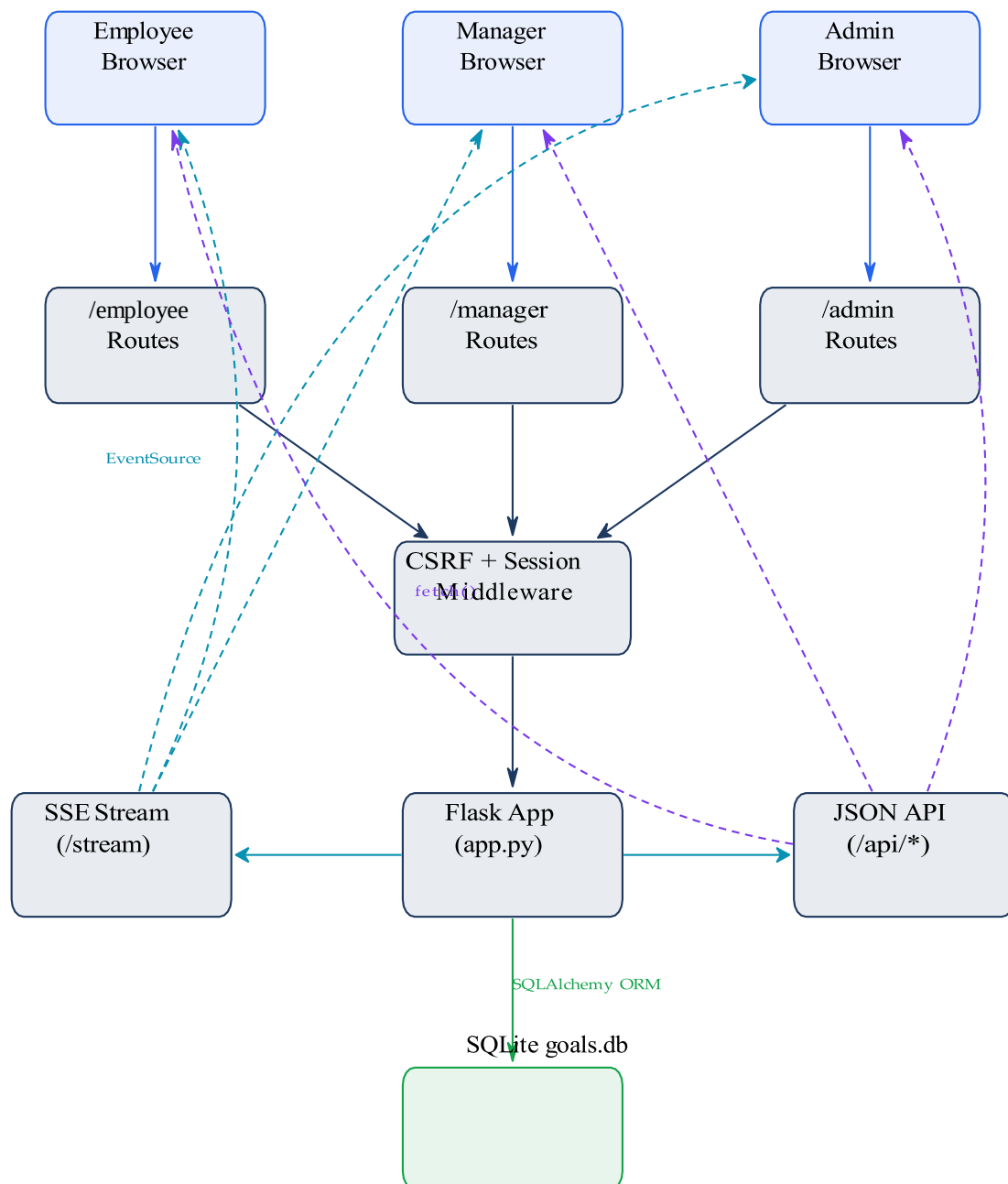
When a manager pushes a shared goal, all employee copies are linked by a shared_goal_group key. Any employee updating their achievement triggers an automatic cascade that updates all linked copies in the same database transaction — ensuring consistency without manual reconciliation.

5.4 Security Architecture

- **CSRF Protection** — Stable per-session token generated at login via secrets.token_hex(); validated with secrets.compare_digest() (timing-safe)
- **Role-Based Access Control** — Every route protected by @role_required() decorator; no bypass path exists
- **Password Hashing** — Werkzeug generate_password_hash() / check_password_hash() throughout
- **Session Hardening** — HTTPONLY, SAME SITE=Lax, production SECURE flag, 7-day lifetime

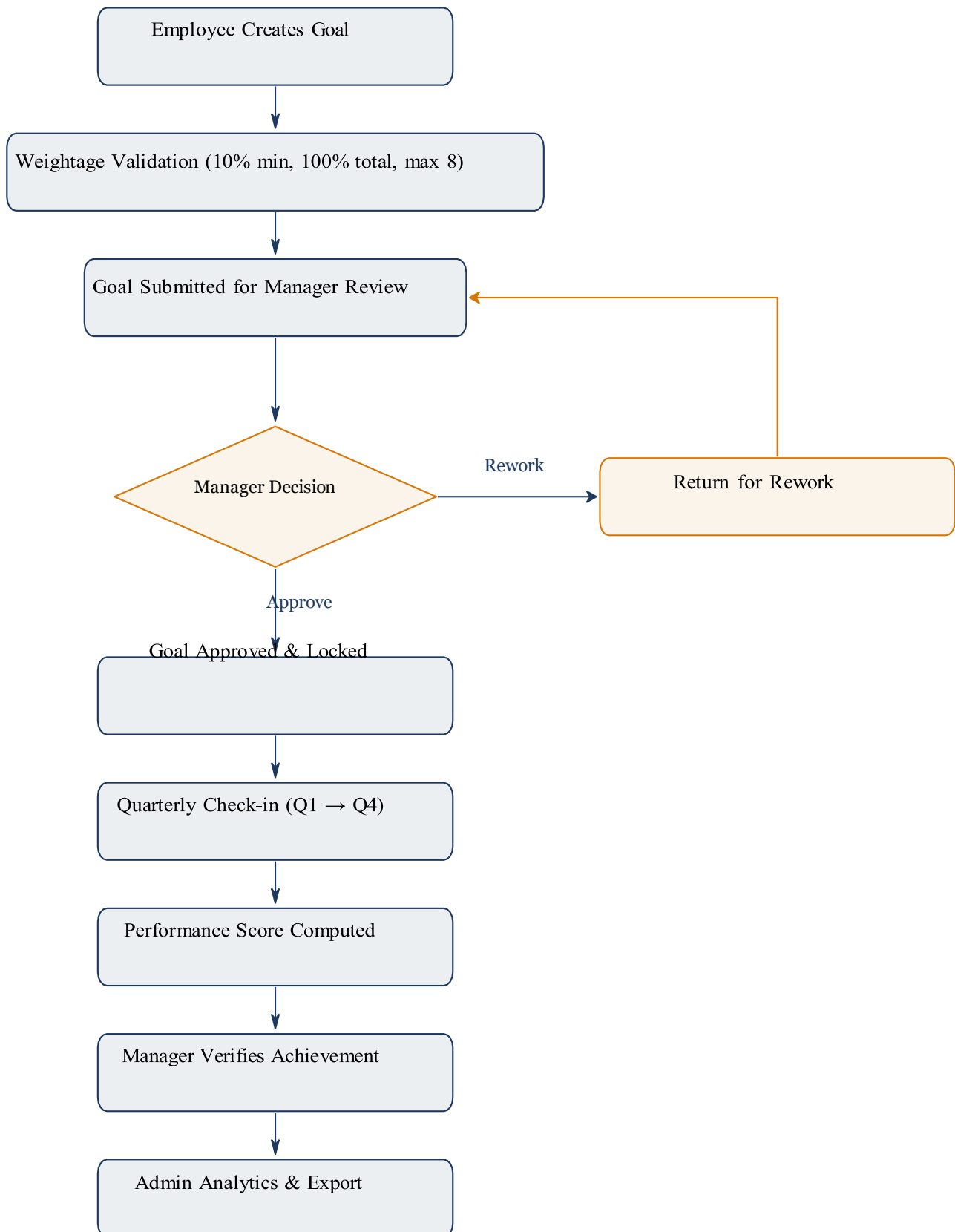
ENTERPRISE SYSTEM ARCHITECTURE

6.1 High-Level Architecture



Blue arrows — HTTP requests from browser to Flask routes **Teal dashed** — SSE push stream (EventSource)
Purple dashed — JSON API fetch (partial DOM patch) **Green solid** — SQLAlchemy ORM reads/writes

6.2 Governance Workflow



PLATFORM INTERFACE PREVIEW

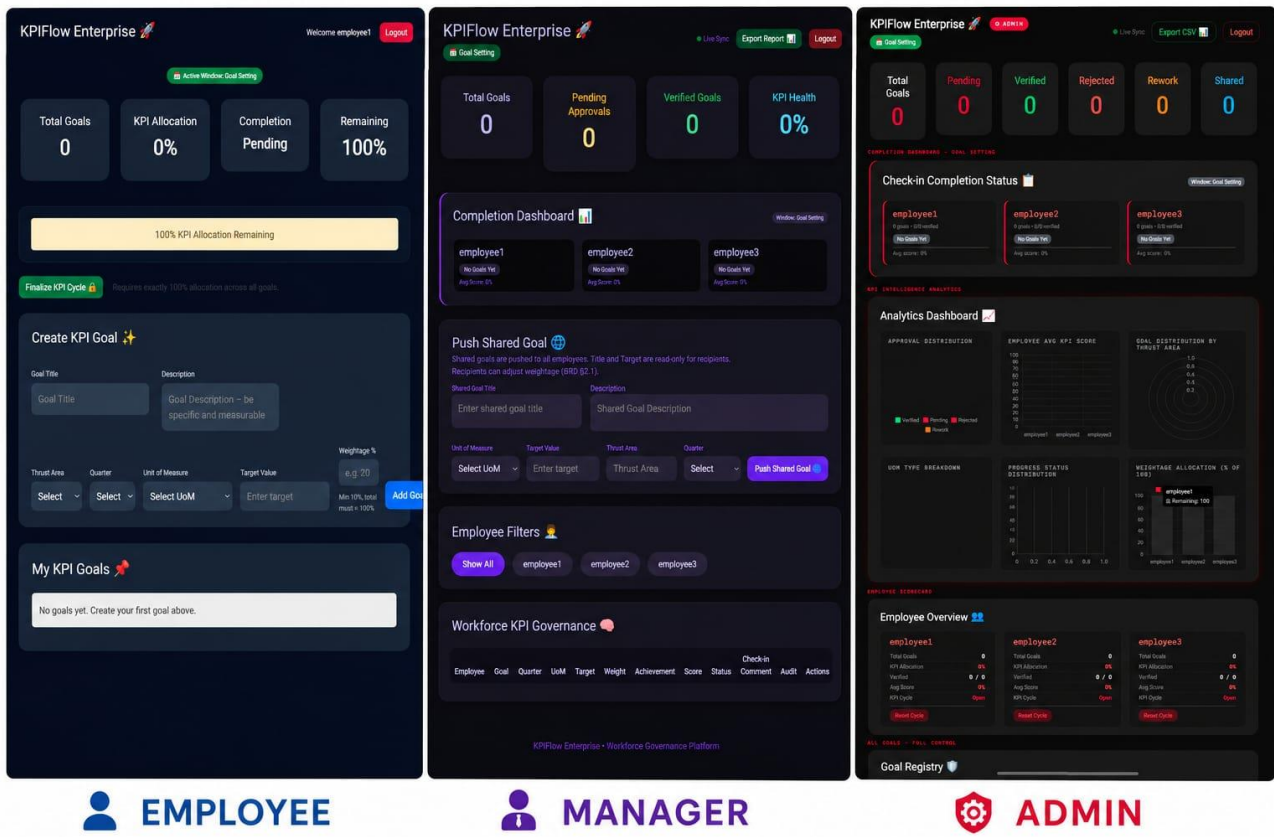


Figure 1: KPIFlow Enterprise role-based dashboard ecosystem

TECHNOLOGY STACK

Layer	Technology & Notes
Backend	Python 3 + Flask — lightweight WSGI; SQLAlchemy ORM for database access
Database	SQLite 3 — zero-config, single-file; auto-created on first run
Auth / Security	Werkzeug password hashing; secrets.token_hex() CSRF tokens; server-side sessions
Frontend	HTML5 + Bootstrap 5.3 (CDN) + Custom CSS; no build toolchain required
Charts	Chart.js 4 (CDN) — 6 chart types, data injected via Jinja2 template context
Real-Time Sync	Server-Sent Events with exponential back-off reconnect
Exports	Python csv stdlib for CSV; openpyxl with styled headers for Excel
Deployment	Render / Railway / PythonAnywhere; single app.py entry point + Procfile

EMPLOYEE DASHBOARD

The Employee Dashboard provides a complete self-service KPI interface with a deep navy/slate design language.

Key Capabilities:

- Create goals with Thrust Area, Quarter, UoM, Target, and Weightage in a single-row inline form
- Real-time KPI allocation meter showing remaining weightage and completion status
- Finalize KPI cycle — locks all goals and prevents further edits without Admin intervention
- Log actual achievement per check-in window (numeric input or date picker for Timeline UoM)
- View performance score, manager comments, and quarterly review notes per goal
- Visual row coding: green border for approved, orange for rework, purple for shared goals
- High Priority badge automatically flags the highest-weighted goal

Design Theme: Deep navy (#020617) background with slate cards (#1e293b), blue accents, and live Live Updated toast on SSE refresh.

MANAGER DASHBOARD

The Manager Dashboard uses a deep purple/dark theme to visually distinguish governance from employee workflows.

Key Capabilities:

- Summary analytics: Total Goals, Pending Approvals, Verified Goals, KPI Health %
- Completion Dashboard — status per employee (Completed / In Progress / Pending) with average score
- Employee filter pills for focused per-employee review
- Per-goal actions: Approve, Verify, Return for Rework, Reject — all with reason capture
- Inline target and weightage editing without leaving the page
- Quarterly check-in comment field saved per goal per window
- Push Shared Goal to all employees simultaneously (with auto-skip for capacity-full employees)
- Live dot indicator showing real-time sync status

Design Theme: Dark purple gradient (#0a0a0f to #12101a) with violet card borders and crimson KPI health indicators.

ADMIN DASHBOARD

The Admin Dashboard uses a black/crimson command-centre theme, reflecting unrestricted system access.

Key Capabilities:

- 6-tile analytics overview: Total, Pending, Verified, Rejected, Rework, Shared goals

- Analytics charts: Approval Distribution (doughnut), Employee Avg Score (bar), Thrust Area breakdown (polar), UoM type breakdown, Progress Status (horizontal bar), Weightage Allocation (stacked bar)
- Employee Scorecard — per-employee goals, weightage, verified count, average score, cycle status with animated progress bar
- Goal Registry — full control table: Force Approve, Unlock, Delete per goal
- Reset KPI cycle per employee
- Escalation Module — flags employees with unsubmitted goals, unfinalised allocations, or missing check-in entries
- CSV and Excel export of all goal data

Design Theme: Pure black (#0a0a0a) with crimson red (#dc143c) accents, JetBrains Mono monospace labels, and a command-terminal visual identity.

UI/UX DESIGN STRATEGY

12.1 Role-Based Visual Identity

Each user role has a deliberately distinct visual language to eliminate ambiguity and reinforce access boundaries:

Role	Primary Colour	Accent	Identity
Employee	Navy #020617	Blue #2563eb	Personal workspace
Manager	Deep purple #0a0a0f	Violet #7c3aed	Governance layer
Admin	Black #0a0a0a	Crimson #dc143c	Command centre

12.2 UX Patterns

- **Row-level status coding** — green (approved), orange (rework), purple (shared), red (rejected)
- **Inline forms** — no modal dialogs; all actions are table-inline to reduce cognitive overhead
- **Dynamic field toggling** — UoM selector hides numeric target and shows date picker when Timeline is selected, with contextual help text per UoM type
- **Flash messaging system** — colour-coded success/warning/danger alerts after every action
- **Live update toast** — blue “*Live updated*” notification fades in/out on SSE refresh
- **Typing guard** — SSE partial refresh is suppressed while the user has a form field in focus

12.3 Quarterly Window Enforcement

Window-gated actions are enforced server-side via `get_active_window()`, with clear UI messaging when a window is closed:

Months	Window	Permitted Actions
May – June	Goal Setting	Create goals; set targets & weightages
July – Sept	Q1 Check-in	Log Q1 achievement; manager comments
October – Dec	Q2 Check-in	Log Q2 achievement; manager comments
January – Feb	Q3 Check-in	Log Q3 achievement; manager comments
March – April	Q4 / Annual	Final achievement entry; year-end review

DEPLOYMENT NOTE

The application is deployed using Render cloud infrastructure. Initial requests may experience brief cold-start delays due to free-tier hosting limitations.

Repository structure:

```

kpiflow-enterprise/
|-- app.py
|-- requirements.txt      # Main Flask application
|-- Procfile              # Flask, SQLAlchemy, openpyxl, Werkzeug
|-- instance/             # gunicorn app:app

|
|-- goals.db              # SQLite database (auto-created)
'-- templates/
    |-- login.html
    |-- employee_dashboard.html
    |-- manager_dashboard.html
    |-- admin_dashboard.html
    '-- escalations.html

```

DEMO CREDENTIALS

Role	Username	Password
Employee	employee1 / employee2 / employee3	pass123
Manager	manager	pass123
Admin	admin	pass123

Live URL: <https://kpiforge.onrender.com/>

Repository: <https://github.com/gautam-gowda/KPIForgee>

FUTURE SCOPE

The following enhancements are identified as the next iteration roadmap:

Enhancement			Description
Microsoft (SSO)	Entra ID		OAuth2 / MSAL integration for single sign-on and org hierarchy sync from Azure AD group membership
Email Notifications			Flask-Mail / SMTP for goal submission, approval, rejection, and check-in reminder events
Microsoft Teams Bot			Adaptive card notifications with deep-link navigation to the relevant goal sheet
QoQ Trend Analytics			Quarter-on-quarter score trend charts per employee and team using historical snapshots
Weighted Score	Aggregate		Per-employee composite KPI score: $\sum \text{Score}_i \times w_i / 100$ displayed on dashboard
Database User Management			SQLAlchemy User model replacing hardcoded credentials; admin user management panel
PDF Goal Sheet Export			weasyprint-based one-page KPI summary PDF per employee
Advanced Filtering			Manager table filter by thrust area, quarter, and progress status via URL parameters

CONCLUSION

KPIFlow Enterprise delivers a complete, production-grade prototype of the Goal Setting & Tracking Portal specified in the AtomQuest Hackathon 1.0 problem statement. The system implements all mandatory Phase 1 and Phase 2 BRD requirements, two bonus governance modules (Escalation and Analytics), and a real-time synchronisation architecture that goes significantly beyond a standard CRUD implementation.

The combination of role-differentiated visual identities, server-enforced business rules, timestamped audit trails, and SSE live dashboards positions KPIFlow Enterprise as a credible foundation for a production internal HR governance tool.

All mandatory BRD requirements: Implemented ✓

Bonus modules: Escalation ✓ Analytics ✓

Real-time sync, CSRF protection, audit trail, export: ✓